

Question				Marking details	Marks Available																					
					AO1	AO2	AO3	Total	Maths	Prac																
3	(d)			<i>Spirogyra</i> – eukaryotic cells & <i>Nostoc</i> - prokaryotic cells (1) Similarity: Both contain ribosomes / cell membranes / DNA/ genetic material (1) Difference: any 2 from: (1 mark) <table><tr><th><i>Spirogyra</i></th><th><i>Nostoc</i></th></tr><tr><td>{Membrane bound organelles/ named organelle} present</td><td>{Membrane bound organelles/ named organelle} absent</td></tr><tr><td>DNA enclosed within nuclear membrane</td><td>DNA free in cytoplasm</td></tr><tr><td>Linear DNA</td><td>Loop of DNA Accept plasmid</td></tr><tr><td>Larger/80s ribosomes Accept ribosomes are different sizes</td><td>Smaller/70s ribosomes</td></tr><tr><td colspan="2">Accept following</td></tr><tr><td>DNA associated with histones</td><td>DNA not associated with histones</td></tr><tr><td>Mesosome absent</td><td>Mesosome present</td></tr></table>	<i>Spirogyra</i>	<i>Nostoc</i>	{Membrane bound organelles/ named organelle} present	{Membrane bound organelles/ named organelle} absent	DNA enclosed within nuclear membrane	DNA free in cytoplasm	Linear DNA	Loop of DNA Accept plasmid	Larger/80s ribosomes Accept ribosomes are different sizes	Smaller/70s ribosomes	Accept following		DNA associated with histones	DNA not associated with histones	Mesosome absent	Mesosome present	1 1 1			3		
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